

Unified Risk Architecture

GLOBAL SCORING ENGINE · US PERIMETER : SYSTEM BLUEPRINT



v.2.4



execution_mode : strict

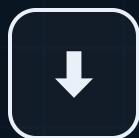


status : active

AUTHORITATIVE MODEL : CRAM-US-2026-07-FREEZE-03

[CORE PIPELINE]

[MIS DISPLAY]



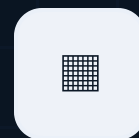
Node 1
Capture



Node 2
DQ_gate



Node 3
normalizeScoreInput



Node 4
scoreCustomer



Node 5
computeGoldenThread



Node 6
computeResidual

Capture → DQ_gate → normalizeScoreInput → scoreCustomer → computeGoldenThread → computeResidual

[NODE 01 → 02] : CAPTURE & DQ TOLLBOOTH

scoreWithDataQualityGate – fail-closed gate before any scoring

STATUS : VALID



PROCEED → **normalizeScoreInput** – composite scoring enabled

STATUS : INVALID



STATUS : HALT · **EXECUTION_TERMINATED** – mandatory field missing / stale screening

GATE CHECKS (FR-007)

- › Customer ID, expected activity, residence/opco & funds geography present
- › Identity source captured (perimeter-scoped – see Node 01_C)
- › Liveness pass unless in-person (branch) channel
- › Screening timestamp fresh (Vital4) – else verification halt

[NODE 01_C] : CAPTURE LAYER · PERIMETER-SCOPED INPUTS

Test Bench capture surface – 8 sections · 36 inputs · all lists verified populating (REV 2.4 fixes)

EXPECTED / OBSERVED ACTIVITY BANDS

TIER	US · GLOBAL_ACCOUNT	UAE · MAL_BANK
1 (Low)	< USD 15k / mo	< AED 50k / mo
2 (Med)	USD 15k–75k / mo	AED 50k–250k / mo
3 (High)	> USD 75k / mo	> AED 250k / mo

source: cramSuiteConfig.expectedUsd / expectedAed – was AED-only, now perimeter-aware ✓

IDENTITY SOURCE (KYC)

PERIMETER	SELECTABLE SOURCES
US · global_account	document · idsp · branch
UAE · mal_bank	uae_pass · emirates_id · idsp · document · branch

uae_pass / emirates_id are UAE-only – now hidden on US ✓ (branch ⇒ liveness waived)

EMPLOYMENT STATUS – KEYED BY LABEL, SCORED BY TIER

Salaried · Pensioner · Student score 1

Self-employed · Freelancer/consultant · Business owner score 2

Unemployed (with activity) score 3

fix: options keyed by unique label (not risk score) – Freelancer/Business owner now display ✓

SECTIONS VERIFIED (36 INPUTS · 0 EMPTY · 0 CONSOLE ERRORS)

- > 1 · Customer Identity

> 2 · Verification Results

> 3 · Geography

> 4 · Business Activity
- > 5 · Products & Use Cases

> 6 · Corridor Risk

> 7 · Screening & Indicators

> 8 · Control Effectiveness

[NODE 04_A] : COMPOSITE EQUATION

Five weighted factors summed; Product/Service & Channel each resolved worst-of first · PEP is a 0% gate

WORST-OF (MAX)

Product/Service ▽
Channel ▽

[3.8]



$\Sigma (\text{FACTOR} \times \text{WEIGHT})$

Geography §7 worst-of + Customer-type + Product/Svc worst-of + Channel worst-of +
Behaviour exp v act

PEP = 0% GATE · screening pre-condition · fail-closed · never summed



LIFECYCLE
WEIGHTS

$\Sigma = \text{decimal}$
composite

US NP-New weights: Customer-type 25% · Geography 25% · Product+Service 20% · Channel 20% · Behaviour 10% · PEP 0% (gate). Weights shift by segment × lifecycle.

[NODE 04_B] : REGIONAL PARAMETER TOGGLE

Geography model diverges by perimeter – US residency-primary worst-of; UAE weighted average

UAE PERIMETER

Geography 20% · Channel 25%

WEIGHTED AVERAGE

Residence 25% · Nationality 20% · Birth 10%

SoW/SoF 15% · Corridor 15% · Digital-Geo 15%

$\Sigma = 100\% \rightarrow$ weighted geography factor

Corridor / Digital-Geo = NULL $\rightarrow$ conservative default Score 2.0 (UAE only)

US PERIMETER

Geography 25% · Channel 20%

WORST-OF (MAX) · residency-primary

✓ Residence (residence / opco) · ✓ SoW · ✓ SoF · ✓ Incorp (entities)

$\rightarrow$ firmToScore $\rightarrow$ clampScore

✗ Nationality – excluded (§11.1 non-discrimination)

✗ Country of birth – excluded (§11.1 · §7.4 Rule 3)

▫ Corridor tier – governance-only (§7.5) · Digital-Geo – not captured (§7.3)

No sub-weights on US – geography is a single worst-of, not a weighted average. Capture bands & identity sources also toggle by perimeter (Node 01_C).

[NODE 04_C] : firmToScore TRANSLATION

Country firm value (0–4) → internal band score; ≥ 4 → Prohibited action

SCORE 1

$\text{firm} \leq 1.5000$

SCORE 2

$1.5 < \text{firm} \leq 2.15$

SCORE 3

$2.15 < \text{firm} < 4.0$

PROHIBITED

$\text{firm} \geq 4.0$

$\text{firm} \leq 1.5 \rightarrow \text{Score 1} \cdot \leq 2.15 \rightarrow \text{Score 2} \cdot \text{else} \rightarrow \text{Score 3} \cdot \geq 4.0 \rightarrow \text{Prohibited}$

OFAC/UN sanctions floors + interim FATF floor + Zenus restricted-country floors are pre-applied in countryRiskPerimeter via lookupCountry().

[NODE 04_D] : OVERRIDE COLLISION ENGINE

PRECEDENCE : PROHIBITED (action) > HIGH > MEDIUM > LOW = math band – three tiers only (Low/Med/High); Moderate-High → High (§2.2/§3.1)

PROHIBITED · action, not a tier · reject / block / exit

OVR-001 confirmed sanctions / TFS true match (SDN·UN·UAE·NACTA) · **OVR-002** prohibited jurisdiction nexus (residence/SoF/SoW/corridor) + Zenus prohibited-restricted country + prohibited business

↳ DPMS: knock-out ONLY if on Zenus prohibited list (A3.5); else DPMS = profession-risk 4 (§5.1), not a knock-out · also: shell bank · TPPP · unregistered MSB · inability to identify UBO

HIGH · mandated floors – equal rank (≤12-mo · 6-mo if MLR0)

OVR-010 confirmed suspicion / existing STR-SAR → +MLR0 · 6-mo review (SAR fired) · **OVR-003** sanctioned-wallet / mixer exposure – any-% direct / tagged / mixer-routed (≥10% ≤5 hops) ↳ direct SDN true-match → Prohibited via OVR-001

OVR-008 Foreign PEP (FATF R.12) · **OVR-009** adverse media · **OVR-004** UBO concealment · **OVR-012** high-risk NOB · **OVR-020** behaviour (device-share / 3+ investigations / pass-through)

OVR-011 Score-3 country exposure – *discretionary* risk-appetite floor (stricter than FATF R.19 Tier-4/5 min; §5.2 25%-weighted input)

MEDIUM **OVR-016** Domestic OR IO PEP – floor only; EDD where the relationship is higher-risk (FATF R.12; FFIEC)

LOW Low/Med/High from the weighted score where no floor fires · raw bands: ≤1.5000 Low · ≤2.1500 Med · >2.1500 High

FINAL_RATING = **MAX_FLOOR** (**MATH_BAND** , **ANY_TRIGGERED_OVERRIDE**) – highest floor wins; Prohibited overrides all

RCA inherits the principal PEP's tier – not a separate floor (FATF R.12) · US: OVR-002/OVR-011 read geoNexusFirmMax = MAX(residence, SoW, SoF, incorp?) – excludes nationality & birth

[NODE 05_5] : GOLDEN THREAD ACTUATION

Inherent rating → controls, approval authority, review cadence, monitoring – inherent drives everything

CDD → EDD

Score 3 → **EDD engaged**

SoF/SoW · adverse media · enhanced
PEP/sanctions

APPROVAL LADDER

RM → CO → **MLRO**

High/EDD → CO/MLRO sign-off

REVIEW CYCLE (MONTHS)

Low 36 · Med 24 · High 12

6 = MLRO-escalated (OVR-010 confirmed
suspicion → 6-mo, SAR fired)

source: policyProfiles.ts (us_baas) · monitoring intensity, sanctions rescreen & adverse-media sweep scale with the inherent band; USD thresholds on US, AED on UAE.

[NODE 06_] : COMPUTE RESIDUAL

Control-adjusted residual band – management information only

RESIDUAL = INHERENT × CONTROL EFFECTIVENESS

Six controls (CDD 20 · SoF 15 · TM 20 · Screening 15 · EDD 15 · Oversight 15)
capped at a one-band step-down · prohibition & override-to-High gates hold

// **FOR MIS DISPLAY ONLY.**
DOES NOT DRIVE INHERENT ACTUATION.

PART D : OPEN ARCHITECTURE & MODULE STATUS

v2.4 · aligned to CRAM-US-2026-07-FREEZE-03 · US perimeter (Zenus rails)

DELIVERED

✓ **Test Bench verified**

8 sections · 36 inputs · 0 empty · risk logic confirmed (Casinos→Prohibited, MSB→High)

✓ **Perimeter-scoped capture**

USD/AED activity bands · US vs UAE identity sources · employment label fix

✓ **Corridor register & UI capture**

9-corridor typology board → TM cards #13/#14 (Global Account)

✓ **Engine Reference**

both CRAMs surfaced – US Mal-CRAM-US-01 v1.0 + CBUAE v1.1

PENDING

Corridor-tier scoring → composite

deliberately governance-only for now (\$7.5)

Digital-geolocation capture

IP / GPS / device geo not yet captured (\$7.3)

OVR-003 wallet-exposure capture field

override inert in engine until captured

Scotermaps · Sub-Tier Approval · UAE 8-Pillar library

exposure mapping · C0 sign-off · country-risk library

SYSTEM_BLUEPRINT_UPDATED v2.4 · WAITING_ON : corridor-tier scoring · UAE library · digital-geo · wallet-exposure capture